

Interfacing Data Transfer Models

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Course Title: Peripherals, Interfacing and Embedded Systems
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Lecture References:

- ▶ **Book:**

- ▶ *Microprocessor Architecture, Programming and Applications with 8085 (Part-III)*, **Author:** Ramesh Gaonkor

- ▶ **Lecture Materials:**

- ▶ *Ramesh Gaonkar*

Peripherals (I/Os) and Interfacing

- ▶ The primary functions of the microprocessor/microcontroller are:
 - ▶ to accept data from input devices such as – keyboards and A/D converters
 - ▶ read instructions from the memory
 - ▶ process data according to the instructions and
 - ▶ send the results to output devices such as – LEDs, printers and monitors.
- ▶ The input and output devices are called either **peripherals or I/O**.
- ▶ Designing logic circuits (hardware) and writing instructions (software) to enable microprocessor to communicate with I/Os is called **interfacing** and logic circuits are called **I/O ports** or **interfacing devices**.

Models of Data Transfer

- ▶ Transmit or receive data occurs either in
- ▶ **Parallel Mode:**
 - ▶ The entire data (i.e., 8-bit or 16-bit) is transferred at one time
 - ▶ In 8085, an 8-bit data is transferred simultaneously over the 8-data lines (i.e., data bus)
 - ▶ Connected via direct cable that has **one wire** for each bit in a character of data code being used by the terminal
 - ▶ With multiple wires, all the bits of a characters can be transmitted between the terminals and computer at once
 - ▶ **Disadv:** Very expensive & no practice over long distance
 - ▶ Devices use parallel mode: *Keyboards, 7-segment Display, Data Converters and Memory.*

Models of Data Transfer

▶ **Serial Mode:**

- ▶ Bits of each character are sent down to a line **one by one**
- ▶ Complicated process because machine needs to know how to **decompose** and to **reconstruct** of bits at each respective end
- ▶ Data are transferred one bit at a time over a single line between the MP and a peripheral.
- ▶ A data is converted to a stream of bits
- ▶ Devices use serial mode: *CRT, Printers, USB, Modems, Telephone lines.*

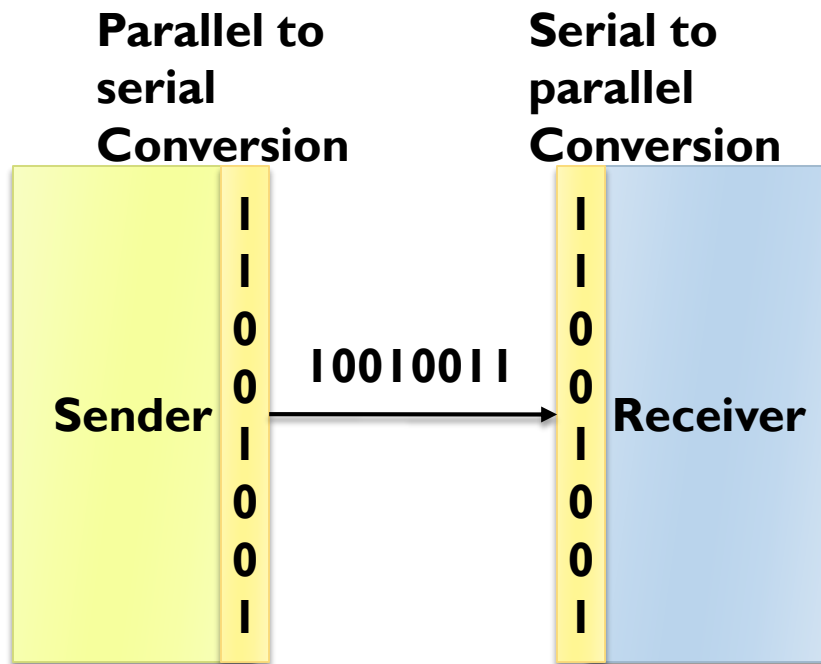
Models of Data Transfer

▶ **Serial Mode:**

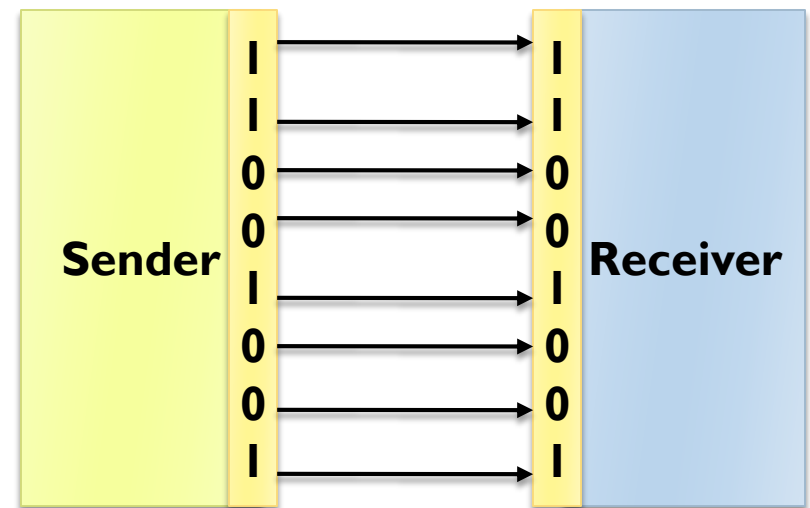
- ▶ RS232- Typical computer COM port
- ▶ SCI- Serial Communication interface, uses the universal asynchronous receiver/transmitter (USART) or UART
- ▶ SPI Serial peripheral interface

Models of Data Transfer

Two basic modes of data transmission



Serial Transmission



Parallel Transmission

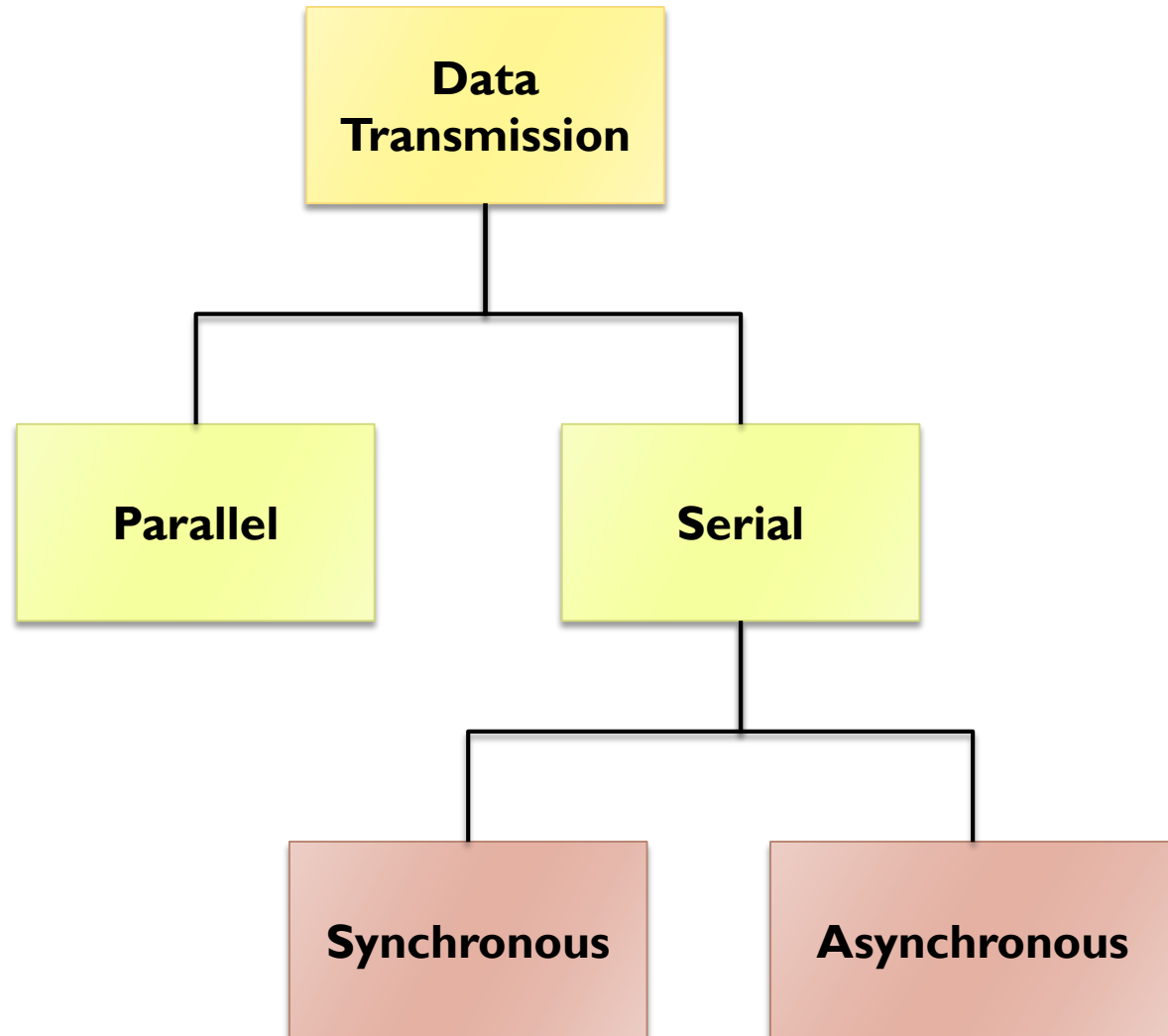
Models of Data Transfer

▶ **Serial**

- ▶ Cheaper
- ▶ Slower

▶ **Parallel**

- ▶ Faster
- ▶ Limited to small distances
- ▶ Simultaneous transmission
- ▶ Requires separate data lines
- ▶ Bits must stay synchronized
- ▶ Expensive



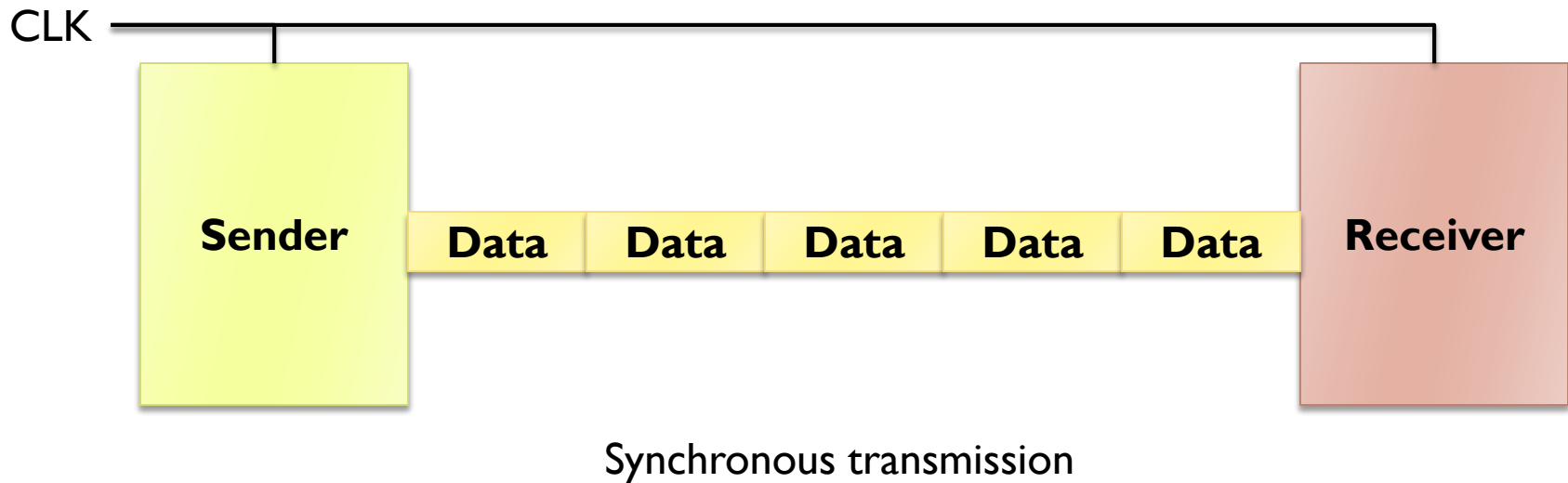
Type of Serial Communication

- ▶ Microprocessor/microcontroller communicates with peripherals in either of two transmission formats:
- ▶ **Synchronous Transmission:**
 - ▶ Synchronous means at the same time
 - ▶ The transmit and receive are synchronized with same clock
 - ▶ Block of data can be sent
 - ▶ It is used for high-speed data transmission

Type of Serial Communication

▶ **Synchronous Transmission:**

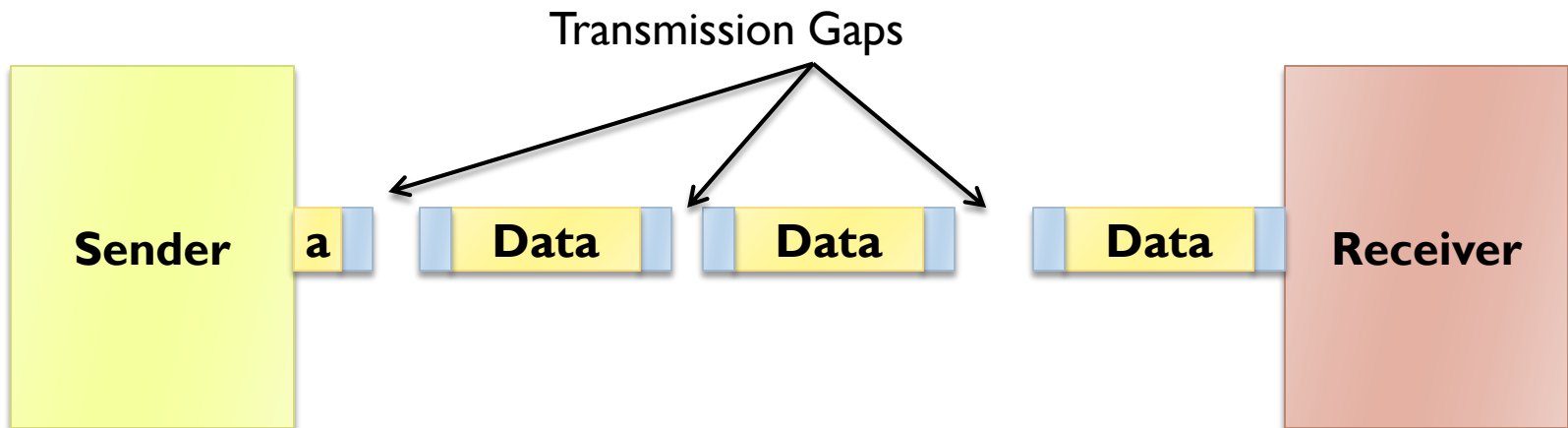
- ▶ Data rates are dependent on clock rates
- ▶ Continuously transmitting characters to remain in sync.
- ▶ More efficient : Less overhead than asynchronous transmission



Type of Serial Communication

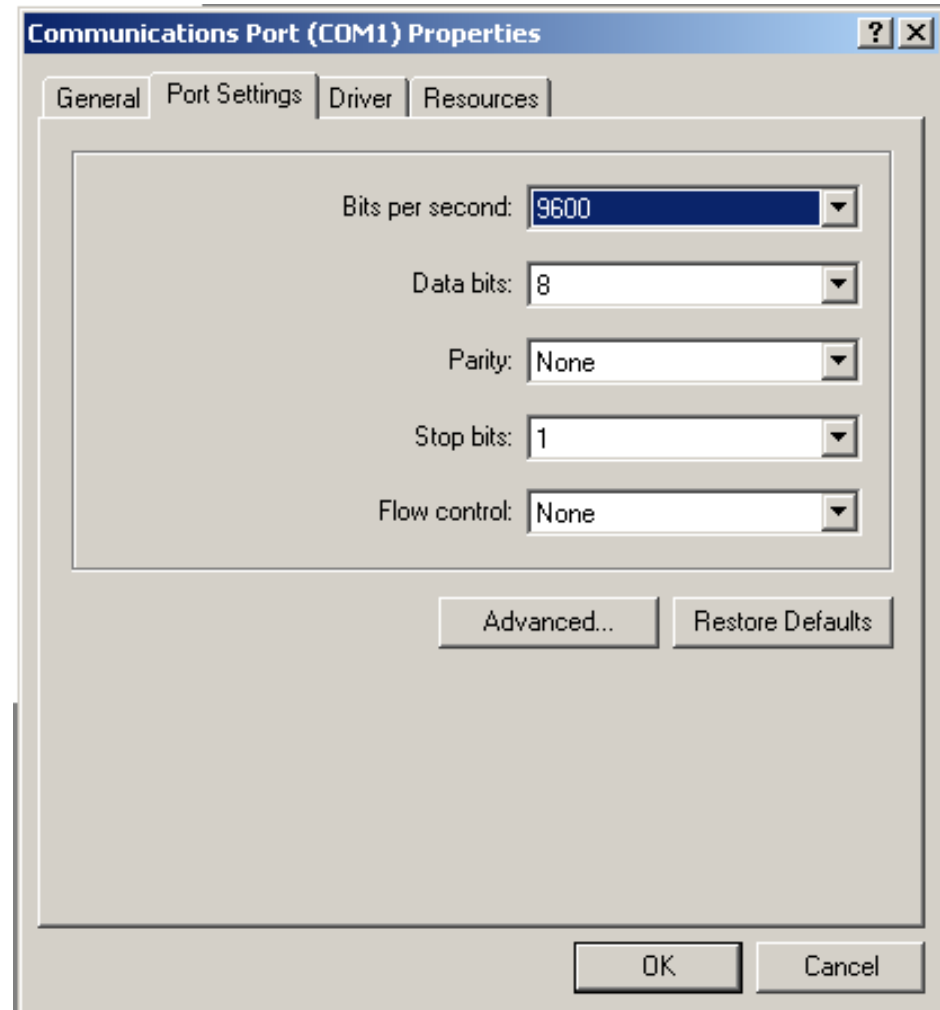
▶ **Asynchronous Transmission:**

- ▶ Asynchronous means irregular intervals
 - ▶ Each byte is encoded for transmission
 - ▶ Start and stop bits
 - ▶ No need for sender and receiver synchronization
 - ▶ It is used for low-speed data transmission
- ▶ Data transfer between microprocessor and peripherals are primarily **asynchronous**.



Asynchronous Serial Transmission

- ◆ Bits are transmitted in a specified format
- ◆ Defined by settings on transmitter and receiver:
 - ▶ Start Bit
 - ▶ Data Bits
 - ▶ Parity Bits
 - ▶ Stop Bits

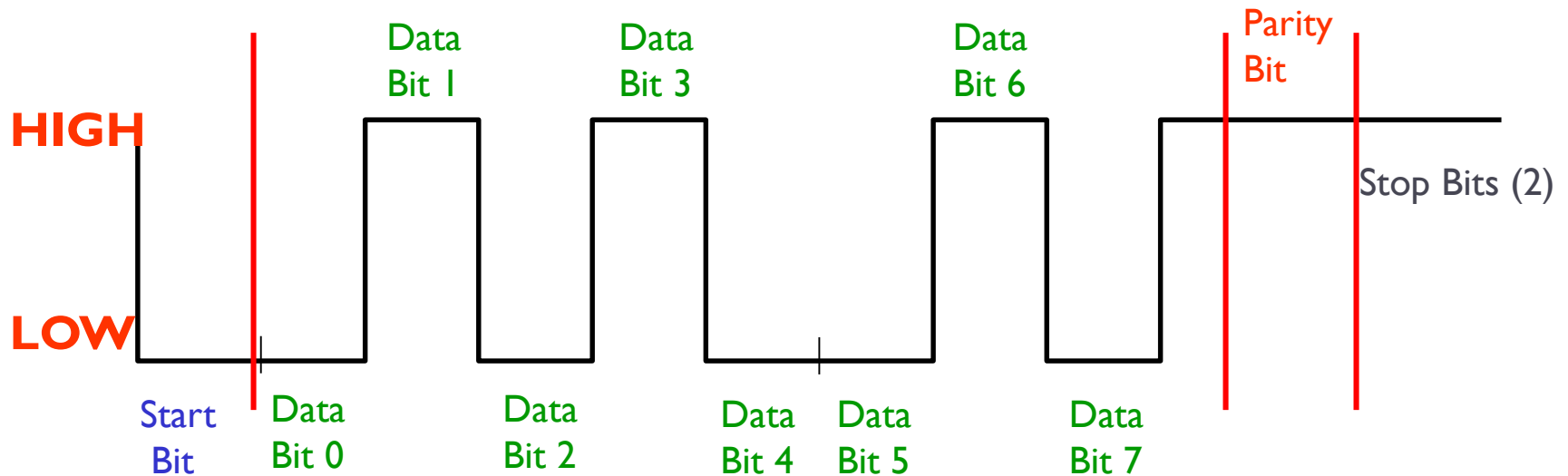


Example of Windows setting

Asynchronous Serial Transmission

One Data Package

◆ Four parts per package



Performance Comparison

- ▶ **Example Scenario:** Consider a character consists of
01001011
- ▶ **Asynch:** $250 \text{ char} \times (8 \text{ data} + 2 \text{ start/stop}) = 2500 \text{ bits}$
- ▶ Assuming 1 **Synch character** after each 50 Character
Transmission **Synch:** $(250 + 5 \text{ synch char}) \times 8 \text{ bits} = 2040 \text{ bits}$
- ▶ Thus, for the scenario Synch is approximately 20% more efficient than Aysnch
- ▶ **Note:** Mostly, host computers adopt Synch transmission.

Conditions for Data Transfer

▶ **Microprocessor Controlled Data Transfer**

▶ ***Unconditional Data Transfer***

- ▶ MP thinks of that a peripheral is always available

▶ ***Data Transfer with Polling***

- ▶ Polling a peripheral or set of peripherals regularly

▶ ***Data Transfer with Interrupt***

- ▶ A peripheral is ready to transfer data and sends an INT signal
- ▶ MP stops current execution and send the peripheral an INTA signal

▶ ***Data Transfer with READY Signal***

- ▶ If peripherals response time is slower than the execution time of MP then T states are extended to complete data transfer

▶ ***Data Transfer with Handshake Signals***

- ▶ Signal are exchanged between MP and peripheral prior to data transfer

Conditions for Data Transfer

▶ **Peripheral Controlled Data Transfer**

- ▶ This type is applied when a peripheral is much faster than the MP
- ▶ In case of **Direct Memory Access (DMA)**, the DMA controller sends **HOLD signal** to the MP and MP releases its Data and Address Bus to the DMA controller
- ▶ Thereupon, data transferred **at high-speed without the intervention of MP**

Thank You !!

